CA2 Requirement Analysis (Project: MyDiet)

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1 Project Description

This project involves the development of a web-based dietary recipe recommendation system designed to support users in making healthier food choices. The system allows users to input personal physical data such as height, weight, and other relevant information, which is used to calculate estimated nutritional requirements. Based on these calculations, the system recommends suitable recipes that meet the user's dietary needs. Additional features include personalised recipe recommendations, a community forum for sharing recipes and experiences, food image recognition to estimate nutritional content, and secure user login functionality.

2 Aims and Objectives

Aims

- To develop a web-based platform that helps users understand their nutritional needs and access suitable dietary recipes.
- To support healthier eating habits through personalised and data-driven recipe recommendations.

Objectives

- User account creation and secure login functionality to store personal profiles.
- Collection of user physical data (e.g. height, weight) and calculation of basic nutritional requirements.
- Generation of recipe recommendations based on calculated nutritional needs.
- Personalised recipe suggestions based on user preferences and previous interactions.
- A recipe forum that allows users to view, share, and discuss recipes.
- Image-based food recognition to estimate the nutritional content of photographed meals.

3 Literature and Background Reading

Existing dietary applications provide useful design references for our system. For example, MyFitnessPal [1] allows users to log meals and automatically track calorie intake through personalised user profiles. We adopt a similar approach by allowing users to enter physical data such as height and weight. This information is then used to calculate nutritional requirements. Unlike traditional tracking apps, our system extends this functionality by providing automated recipe recommendations instead of relying solely on manual food logging.

Research has shown that monitoring dietary behaviour and nutritional indicators improves users' awareness of healthy habits and supports behavioural change [2]. Therefore, BMI and calorie requirement calculations are incorporated into the system. These calculations provide personalised, data-driven dietary guidance tailored to individual users.

Studies on food recommender systems, such as the PREfER system [3], demonstrate that personalised recommendation algorithms increase user engagement compared with static recipe databases. Rather than requiring users to search for meals manually, recommendation systems can automatically suggest suitable options. Based on these findings, our system implements adaptive recipe suggestions. These recommendations respond to user preferences and interaction history, allowing more relevant meal suggestions over time.

Recent advances in computer vision enable automatic dietary assessment through food image recognition. Deep learning models have been shown to accurately identify food items and estimate nutritional values from images [4], [5]. Manual food entry can be time-consuming and prone to error. Image-based estimation reduces user effort while improving usability. For this reason, our system includes an image recognition feature that allows users to photograph meals and receive estimated nutritional information automatically.

Behavioural research highlights the importance of social interaction in supporting long-term lifestyle change. Studies of online health communities show that peer discussion and shared experiences increase motivation and engagement [6]. Similarly, smartphone-based support groups have been shown to improve adherence to healthy eating behaviours in daily life [7]. Based on these findings, our system incorporates a recipe forum. This forum allows users to share recipes, exchange experiences, and encourage one another.

Because the system processes personal and health-related data, privacy and security are essential design considerations. The General Data Protection Regulation (GDPR) establishes requirements for secure personal data processing and informed user consent [8]. In addition, OWASP security guidelines identify common web vulnerabilities such as SQL injection and broken authentication, and recommend mitigation techniques including password hashing and secure session management [9]. Guided by these principles, our system implements secure login functionality and protected data storage mechanisms.

4 Development & Implementation Summary

The MyDiet system is architected as a modular web application consisting of three distinct layers: a reactive Frontend Client, a central RESTful API Server, and a specialized AI Microservice. The Frontend Client handles all user interactions across the five core modules (Homepage, Plan, Identifier, Community, Profile). It communicates with the API Server via asynchronous HTTP requests to retrieve personalized meal plans and synchronize user data, such as daily caloric intake and social posts. A key architectural decision is the isolation of the 'Identifier' module: when a user uploads a meal photo, the main server acts as a proxy, forwarding the image data to the AI Microservice. This service processes the image and returns nutritional estimates back through the chain, ensuring the main application thread remains unblocked during heavy computational tasks.

To implement this design, we selected the PERN stack (PostgreSQL, Express, React, Node.js) supplemented by Python.

- **Frontend:** We chose React.js for its component-based architecture, which allows us to reuse UI elements like the 'Daily Intake' component across different pages.
- **Backend & Database:** Node.js and Express were selected for their non-blocking I/O, which is ideal for handling real-time community interactions. For storage, we chose PostgreSQL over NoSQL alternatives. Since our "Plan" module relies on strict relationships between user biometrics, ingredients, and weekly schedules, a relational database ensures data integrity and complex querying capabilities.

- **AI Service:** The image recognition component will be built using Python. We chose Python here simply because it offers the most robust support for machine learning libraries (e.g., TensorFlow/PyTorch), allowing us to implement the food recognition feature more efficiently than if we attempted it in JavaScript.

To manage the project within the ten-week constraint, we adhere to an Agile methodology using Git for version control. The workflow is structured into three distinct phases. We begin with a rigorous System UI Design phase (Feb 17 – Mar 12), finalizing the database schema and wireframes to ensure a stable foundation before coding begins. Core Development launches first on Feb 17, prioritizing the Profile module and the Weekly Meal Plan algorithm. As these core features stabilize, we will initiate the Community Feature Development in March, allowing the two streams to proceed in parallel. The final stage is a dedicated Integration Testing period (Apr 13 – Apr 26), where frontend and backend modules are merged, and bug fixing is conducted prior to the final demonstration.

5 Data Sources

This project involves processing publicly available text and utilizing datasets from the public domain to generate nutritional insights. As outlined below, we have selected three specific data sources to ensure the system is populated with accurate, scientifically validated information.

- **Recipe Data: Food.com Dataset [10]**
  To provide a comprehensive library for recipe discovery, we are using a dataset from the public domain hosted on Kaggle, originally sourced from Food.com. This dataset contains over 230,000 entries, including metadata such as ingredient lists and pre-calculated nutritional profiles.
  - **Acquisition:** The data will be obtained as static CSV files (RAW_recipes.csv) directly from the platform. A Java-based parser will process this data to populate our local SQLite database.
  - **Permission:** We confirm we are using it with permission, as the dataset is released under a license permitting educational and research use.
- **Ingredient Data: CoFID (UK Standard) [11]**
  For the specific function of tracking raw ingredients (e.g., logging "100g of raw chicken breast" rather than a full meal), we will utilise McCance and Widdowson's The Composition of Foods Integrated Dataset (CoFID).
  - **Relevance:** Using this UK-specific data ensures that when users are tracking their raw ingredient stocks, the nutritional calculations reflect local food standards.
  - **Permission:** This data is published under the Open Government Licence (OGL), which allows for free use and adaptation.
- **Nutritional Standards: NCBI Dietary Reference Intakes (DRI) [12]**
  To evaluate the user's diet, the system requires a baseline for nutritional needs. We will use the Dietary Reference Intakes (DRI) tables from the NCBI Bookshelf.
  - **Acquisition:** We will extract data from the Reference Tables for macronutrients, vitamins, and elements to create a structured reference table within the application.
  - **Permission:** As a publication of the US Federal Government, these tables are in the public domain and available for unrestricted use.
- **Confidentiality of Personal Information**
  The system will generate user-specific data, including dietary logs and raw ingredient tracking history.
  - **Ethical Compliance:** To ensure the confidentiality of any personal information, all data will be securely stored in our database with strict access controls. We will apply data minimisation by collecting only essential login details, avoiding sensitive identifiers like real names or emails. Furthermore, passwords will be cryptographically hashed (e.g., SHA-256) and never stored in plaintext.

6 Testing and Evaluation

To ensure accuracy, reliability, and user safety, the system will undergo rigorous technical testing and user evaluation.

Technical & Performance Testing

- **Algorithm Validation (Unit Testing):** Using JUnit and Mockito frameworks, unit tests will rigorously evaluate the Java-based dietary recommendation algorithm against the NCBI Dietary Reference Intakes (DRI) baseline. This ensures the logic never generates clinically unsafe or extreme diet plans.
- **Integration & Security Testing:** API endpoints built with Spring Boot will undergo integration testing to verify the "Local-First" architecture. Tests will confirm that special category data (e.g., weight, allergies) remains locally encrypted and is not inadvertently exposed or stored unencrypted on the backend database.

System Evaluation (Beta Testing)

To evaluate whether the system meets the objective of supporting healthier eating habits, we will conduct an evaluation phase utilizing external beta testers. Participants will use the MyDiet application to track meals and generate plans, followed by completing a structured questionnaire. This survey will assess specific objectives: UI/UX accessibility, the accuracy of the image recognition feature, and the perceived relevance of the personalised recipe recommendations.

7 Ethical Considerations

This system processes personal and health-related information and provides automated dietary recommendations. Therefore, ethical responsibility is embedded in its design to ensure user safety, privacy, and responsible community interaction.

1. **Medical Safety and Non-Maleficence**
   Since the software offers dietary planning, there is a risk that users might mistake the automated suggestions for professional medical advice. Unsuitable recommendations for users with existing health issues (such as diabetes, hypertension, or allergies) could cause physical harm. To mitigate this, we will ensure that:
   - A clear disclaimer appears during the sign-up process and before any meal plan is created. It will state that the software offers general information only and is not a substitute for medical diagnosis or treatment.
   - The software's logic is strictly limited to nutritional planning. It will not attempt to diagnose illnesses or assess clinical risks.
   - The software will not generate diet plans that are extreme or clinically unsafe (e.g., starvation diets).
   - All advice is based on established nutritional standards to guarantee scientific accuracy.
2. **Data Privacy and Protection**

The system needs to handle sensitive data, including user weight, food allergies, and health history. This information is classified as "special category data" under UK law. We will strictly follow the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018. Specifically, we will:

- Practice data minimisation by asking only for the details truly needed to create a diet plan.
- Get clear permission (explicit consent) from users before saving any health-related data.
- Give users full control, allowing them to view, download, or delete their data at any time.
- Protect data security by using encryption when data is being sent or stored.
- Never sell or share private health details with advertisers or third parties.
- Perform regular security checks to prevent data leaks or hacking.

3. **Community Standards and Safety**
The built-in community section allows users to share tips and experiences. However, this creates risks regarding false claims, bad health advice, harassment, or content promoting self-harm. We will ensure that:
   - Users must agree to a specific Community Code of Conduct when they register.
   - This Code will strictly forbid posts about extreme fasting, self-harm, unproven medical cures, illegal drugs, or bullying.
   - A zero-tolerance rule applies to serious violations, leading to immediate bans.
   - Users have an easy way to report content that is harmful or misleading.
   - Posts made by users are visually distinct from the software's official, evidence-based advice.
   - While users are responsible for what they post, the platform will actively monitor the space to keep it safe.
4. **Scope and Cultural Flexibility**
The software is built to offer healthy eating advice based on general nutritional science. We acknowledge that we must respect different eating habits, even though the system cannot automatically cater to every specific cuisine or religion. We will ensure that:
   - The system is transparent about its limitations, stating that recommendations are based on general nutrition rather than specific cultural rules.
   - The software allows users to change or swap out suggested meals to fit their personal, religious, or cultural needs (e.g., Halal, Kosher, or Vegetarian).
   - The algorithm does not present one specific cultural diet as the only "healthy" option.
   - The system avoids rigid "ideal" models, offering flexible options that stay within healthy nutritional limits.

8 BCS Project Criteria

The development of MyDiet demonstrates strict adherence to the professional and ethical standards established by the BCS.

Legal and Regulatory Compliance: The project strictly complies with the UK GDPR and Data Protection Act 2018. The system demonstrates this by practicing data minimisation, requiring explicit consent before processing biometric data, and leveraging Java's robust security frameworks (e.g., Spring Security) to ensure sensitive health information remains encrypted and strictly controlled.

Public Interest and Medical Safety: Recognizing the risks associated with automated dietary advice, the system prioritises user safety. It supports public health by enforcing strict backend algorithms that prevent the generation of clinically unsafe diets. Furthermore, clear disclaimers are embedded within the UI to ensure users do not mistake automated nutritional planning for professional medical diagnosis.

Professionalism and Social Responsibility: The system demonstrates professional integrity by implementing a strict Community Code of Conduct to mitigate risks such as the promotion of self-harm or extreme fasting within the forum. Additionally, the use of enterprise-grade technologies (Java/Spring Boot), test-driven approaches (JUnit), and Agile methodologies highlights a highly professional approach to software engineering.

9 UI / UX Mockup

The MyDiet application comprises five primary functional modules: Homepage, Plan, Identifier, Community and Profile.

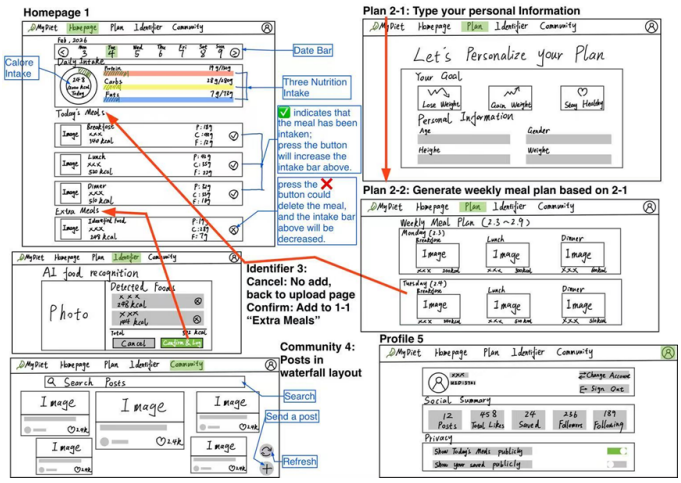

Figure 1: MyDiet Application UI Mockup

After registration, users are initially directed to Interface 2-1 to provide their personal information. Following data submission, the system transitions to Interface 2-2, where a customized Weekly Meal Plan is algorithmically generated based on the user's biometrical data. Interface 1 features a Daily Intake component; users can modulate values of the component via the Today's Meals or Extra Meals sections. Furthermore, a Date Bar functionality is integrated to facilitate the retrieval and review of historical intake data.

After food photo uploaded and AI-driven analysis, Interface 3 will display the nutritional specifications. Selecting "Cancel" reverts the user to the upload page, while "Confirm" integrates the identified item into the Extra Meals section of Interface 1. Interface 4 serves as a community platform, enabling users to publish, browse, and search for diet-related content. Finally, Interface 5 is the personal information page, where you can view the Social Summary and adjust Privacy.

10 Project Plan & Risks

COMP208 Project

Name of Project	MyDiet	Start Date	2026-01-26
Team Members	Zihan Ding, Liming Li, Xiang Lin, Liangyu Liu, Qiyu Mao, Yiming Wang, Tian Yu	End Date	2026-05-18

WBS ID	Task Name	Task Owner	Start Date	End Date	Duration (Days)	Progress (%)
1	**Project Initiation & Requirement Analysis**					
1.1	Team introduction and ice-breaking activities	Zihan Ding, Liming Li, Xiang Lin, Liangyu Liu, Qiyu Mao, Yiming Wang, Tian Yu	2026-01-26	2026-01-26	1	100%
1.2	Discuss & Define Project Scope	Zihan Ding, Liming Li, Xiang Lin, Liangyu Liu, Qiyu Mao, Yiming Wang, Tian Yu	2026-01-26	2026-02-08	10	100%
1.3.1	Project Description and Aims and Objectives	Liangyu Liu	2026-02-08	2026-02-15	5	100%
1.3.2	Development and Implementation Summary & Testing and Evaluation & BCS Project Criteria	Liming Li	2026-02-08	2026-02-15	5	100%
1.3.3	Key Literature and Background Reading & References	Tian Yu	2026-02-08	2026-02-15	5	100%
1.3.4	Data Sources & References	Qiyu Mao	2026-02-08	2026-02-15	5	100%
1.3.5	UI / UX Mockup	Yiming Wang	2026-02-08	2026-02-15	5	100%
1.3.6	Project Plan (including Risks and Contingency Plans) & Ethical Considerations	Zihan Ding, Xiang Lin	2026-02-08	2026-02-15	5	100%
1.4	Document Integration	Zihan Ding, Liming Li, Xiang Lin, Liangyu Liu, Qiyu Mao, Yiming Wang, Tian Yu	2026-02-15	2026-02-17	2	100%
2	**System & UI Design**					
2.1	System architecture and database design	Qiyu Mao	2026-02-17	2026-03-03	11	0%
2.2	UI wireframe and prototype design	Yiming Wang	2026-02-17	2026-03-03	11	0%
2.3	Complete design document	Qiyu Mao,Yiming Wang	2026-03-04	2026-03-12	7	0%
3	**Core Development**					
3.1	Profile module	Zihan Ding, Liming Li	2026-02-17	2026-02-26	7	0%
3.2	Weekly Meal Plan algorithm development	Xiang Lin, Liangyu Liu, Tian Yu	2026-02-17	2026-03-31	12	0%
3.3.1	Front-end development (diet planning)	Zihan Ding, Liming Li	2026-02-27	2026-03-11	10	0%
3.3.2	Back-end development (diet planning)	Xiang Lin, Liangyu Liu, Tian Yu	2026-03-02	2026-03-11	7	0%
4	**Community Feature Development**					
4.1	Community feature design	Qiyu Mao,Yiming Wang	2026-03-09	2026-03-20	10	0%
4.2	Community post and sharing implementation	Zihan Ding, Liming Li	2026-03-09	2026-03-20	10	0%
4.3	Community interaction and moderation	Xiang Lin, Liangyu Liu, Tian Yu	2026-03-12	2026-03-20	7	0%
5	**Integration & Testing**					
5.1	System integration	Qiyu Mao,Yiming Wang，Zihan Ding	2026-04-13	2026-04-26	10	0%
5.2	Testing and bug fixing	Liming Li,Xiang Lin, Liangyu Liu, Tian Yu	2026-04-13	2026-04-26	10	0%
6	**Documentation & Demonstration**					
6.1	Demo Documentation	Qiyu Mao,Yiming Wang，Zihan Ding	2026-04-27	2026-05-04	6	0%
6.2.1	Individual Submission	Liming Li,Xiang Lin	2026-04-27	2026-05-05	7	0%
6.2.2	Final documentation and project submission	Liangyu Liu, Tian Yu	2026-05-06	2026-05-10	4	0%
7	**CA2 Requirement Analysis**					
6.2.2	CA2 Requirement Analysis	Zihan Ding, Liming Li, Xiang Lin, Liangyu Liu, Qiyu Mao, Yiming Wang, Tian Yu	2026-01-26	2026-05-10	104	25%

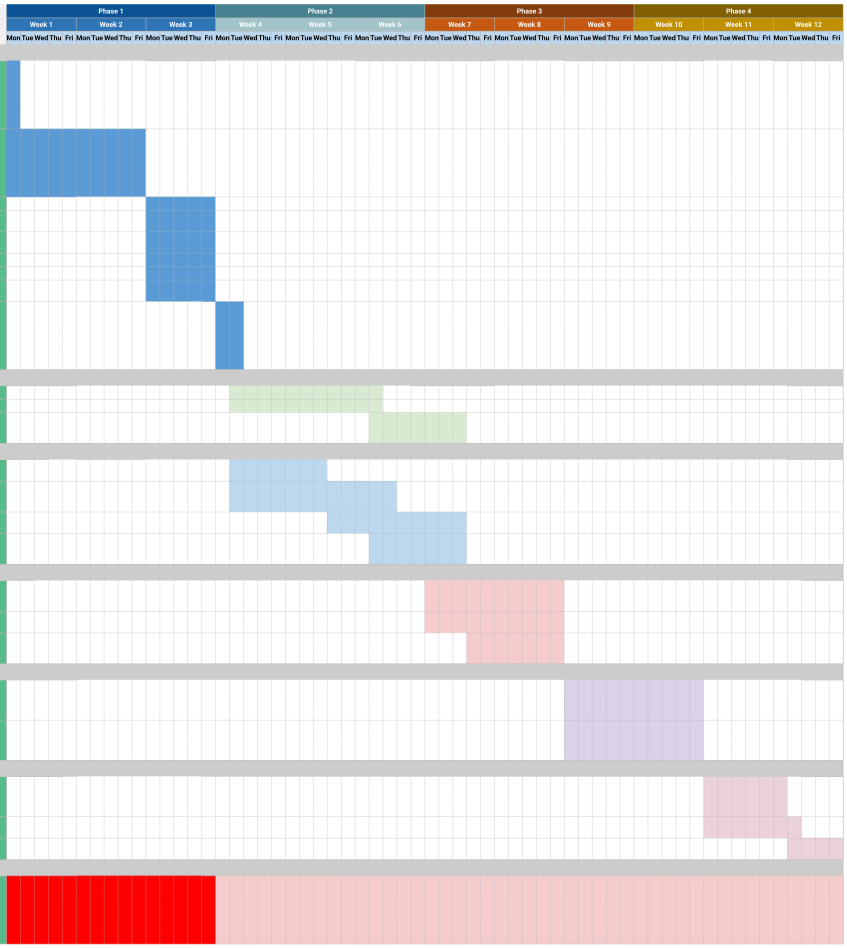

Figure 2: Project Gantt Chart

Risks	Contingencies	Likelihood	Impact
Hardware Failure Damage to a team member's laptop or a development environment crash, resulting in the loss of local code.	**Prevention:** Enforce strict version control (Git) and require daily code pushes to a cloud repository (e.g., GitHub/GitLab). **Handling:** Maintain clear documentation for the development environment setup. If personal hardware fails, use university lab computers to quickly reconfigure the environment and continue work.	Low	High
Software & Programming Failure Critical bugs in core features (e.g., calorie calculation algorithms) or issues with the nutrition database API (inaccuracy/downtime).	**Prevention:** Implement Unit Testing; configure timeout protection and error handling when calling external nutrition data APIs. **Handling:** Prepare a local set of "static mock data" as a backup. If the external API fails during the demo, switch to the mock data to ensure the application flow remains functional.	Medium	High
Running Out of Time Development time being compressed due to conflict with midterms/finals or heavy workload from other courses.	**Prevention:** Create a Gantt Chart with buffer periods at the start of the semester. **Handling:** If behind schedule near the deadline, decisively cut "nice-to-have" features (e.g., social sharing, AI recipe generation) to ensure core functions are delivered.	High	High
Scope Creep Continuous addition of new features (e.g., adding an e-store or video tutorials) causing the team to lose focus and fail to complete the code.	**Prevention:** Freeze the requirements document. Any new feature must be voted on by the team and assessed for its impact on the deadline. **Handling:** Move new ideas to a "Future Iteration List" to be included in the written report as future outlooks, but exclude them from the current implementation.	High	Medium
People Risks Team members falling ill, becoming unresponsive due to other commitments, or internal conflicts regarding technical choices.	**Prevention:** Implement "Pair Programming" to facilitate knowledge sharing and avoid single points of failure; hold brief weekly sync meetings. **Handling:** If a member is unable to complete tasks, the team lead must reassign the workload immediately or seek guidance from the course tutor/supervisor.	Medium	High

Table 1: Project Risk Assessment

11 References

[1] MyFitnessPal Inc., "MyFitnessPal: Calorie Counter and Diet Tracker," [Online]. Available: https://www.myfitnesspal.com. [Accessed: Feb. 18, 2026].

[2] A. Chatterjee, R. Bajpai, and M. W. Gerdes, 'Analyze the Impact of Healthy Behavior on Weight Change with a Mathematical Model using the Harris-Benedict Equations', Jun. 03, 2021, In Review. doi: 10.21203/rs.3.rs-584141/v1.

[3] D. Bianchini, V. De Antonellis, N. De Franceschi, and M. Melchiori, 'PREFer: A prescription-based food recommender system', Computer Standards & Interfaces, vol. 54, pp. 64–75, Nov. 2017.

[4] R. U. Haque, R. H. Khan, A. S. M. Shihavuddin, M. M. M. Syeed, and M. F. Uddin, 'Lightweight and Parameter-Optimized Real-Time Food Calorie Estimation from Images Using CNN-Based Approach', Applied Sciences, vol. 12, no. 19, p. 9733, Sep. 2022.

[5] P. N, S. Poojitha, S. M. Hussain, M. P. Chowdary, and M. Rafi, 'Food Recognition and Calorie Estimation in Mixed Food Items using MobileNet', in 2025 6th ICICV, IEEE, Jun. 2025, pp. 1014–1019.

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[12] Institute of Medicine (US) Committee (2011) 'Dietary Reference Intakes for Adequacy', in Dietary Reference Intakes for Calcium and Vitamin D.